GEOMETRICALLY ORDERED DYNAMICS
FOLIO IT-OMEGA



OmegaCDM

A Pre-Registered Cosmological Prediction ($\Omega\Lambda = \ln 2$)

Yett, R.W.

Independent Research, Arkansas

RESULTS STATED IN TIER — LIMITS FIRST

CONVERGENCE -- INFORMATION TENSION

This folio converges on **information tension** from the cosmological preregistration angle: the one place where the law's constants are staked against sky data with pre-declared kill criteria, a runtime verification script, and no hardcoded verdicts. Its unique angle is the **preregistration structure itself** — the prediction $\Omega_\Lambda = \ln 2$, the precision ladder that would kill it, and the internal tensions stated against the author's own interest. Sibling folios approach the same claim from other angles: [TIER: CONJECTURE] the pentagons folio (IT-PLEROMA) takes the axiomatic angle, stating the law whose constants are tested here. Limits first: the prediction was never actually registered; the dark-energy program it belonged to is closed and ΛCDM stands; the Ω contradiction ($1/\sqrt{2}$ vs $\ln 2$) is open, not resolved. This paper is a status record, not a claim of victory.

SUPERSESSION NOTE

This folio **supersedes** Zenodo record 21867985 (10.5281/zenodo.21867985), *ΩCDM: A Pre-Registered Cosmological Prediction:* $\Omega_\Lambda = \ln 2$, per the canon rule: the correction *becomes* the paper. The superseded record is retained as history. What changes: the original record's preregistration framing is kept, its runtime verification script is shipped alongside and excerpted in Appendix A, and a status note is added (§4) recording that the dark-energy front is dead, the preregistration was never pursued, record 30's $\Omega_\Lambda = 1/\sqrt{2}$ is withdrawn as circular, and the Ω contradiction remains open. No resolution is invented; no new numbers are introduced. Every number below is recomputed at runtime by the shipped script.

1 The prediction

The claim, in the exact form the verification script checks it. Inputs are published values, cited inline; change them and the conclusions move. The constant:

$$\Omega_\Lambda = \ln 2 = 0.6931472. \tag{1}$$

P1: the Hubble constant. Flatness plus (1) gives $\Omega_m = 1 - \ln 2 = 0.306853$. With the physical matter density $\omega_m = \Omega_m h^2 = 0.1430$ (Planck 2018),

$$h = \sqrt{\frac{0.1430}{1 - \ln 2}} = 0.682658, \qquad H_0 = 68.27 \text{ km s}^{-1} \text{ Mpc}^{-1}. \tag{2}$$

For comparison: Planck 67.36; SH0ES 73.04 ± 1.04 (Riess et al. 2022) — a -4.59σ tension with the registered value, stated here because the preregistration demands it.

P2: the matter density. $\Omega_m = 0.306853$ against Planck's 0.3153 ± 0.0073: $+1.16\sigma$. Consistent.

The assumed length choice. The chain closes only on a stated assumption: with the curvature radius $L = \Lambda^{-1/2}$ the prediction lands on $\ln 2$; with the Hubble radius $L = c/H_0$ it lands on $\ln 2/3 \approx 0.231$, excluded by observation. The length choice is the assumption that makes the chain close, and it is labeled as such — not derived.

2 Retrodiction status: why it was not registered

Against Planck 2018 ($\Omega_\Lambda = 0.6847 \pm 0.0073$), the $\ln 2$ value sits at $+1.16\sigma$: consistent at Planck precision, **and known before filing**. It was therefore a postdiction, and it was correctly withheld from registration. The script prints this verdict from a comparison it performs; no verdict in the file is hardcoded. This section exists because the preregistration methodology requires anti-backdating: a prediction known before filing cannot be registered afterward.

3 Kill thresholds

A preregistration is worth what would kill it. Three are named; each is computed, not asserted.

The precision ladder (P3). If the measured center stays at the Planck value while precision improves, the 3σ window around $\ln 2$ eventually excludes it:

$\sigma(\Omega_\Lambda)$	3σ window around $\ln 2$	status
0.0073 (Planck 2018)	[0.6712, 0.7150]	alive (1.16σ)
0.0040 (DESI+Euclid)	[0.6811, 0.7051]	alive (2.11σ)
0.0028 (CMB-S4 class)	[0.6847, 0.7015]	**falsified** (3.02σ)

The kill threshold is reached at $\sigma(\Omega_\Lambda) = 0.0028$.

The internal tension, stated against our own interest. At the SH0ES value $H_0 = 73.04$, $\Omega_\Lambda = 1 - \omega_m/h^2 = 0.7320$, which is 5.3σ (Planck error bar) from $\ln 2$. Meanwhile $a_0 = cH_0/2\pi$ prefers $H_0 \sim 73.04$. The two are **not jointly satisfiable**: $\Omega_\Lambda = \ln 2$ needs $H_0 \sim 68.3$. A converged H_0 kills at least one.

The epoch window. The coincidence problem is unresolved: $\Omega_\Lambda(z)$ stays within 1.23% of $\ln 2$ only for $-0.0262 \leq z \leq +0.0001$, a window of $\Delta z = 0.0263$. OmegaCDM selects an epoch, not just a value.

4 Status note

This is the fix the audit index required, stated plainly:

- *The dark-energy front is dead.* ΛCDM stands. This paper makes no claim that $\Omega_\Lambda = \ln 2$ replaces ΛCDM physics, reopens the dark-energy program, or carries any derivation of cosmic acceleration. The DE program is closed.
- *The preregistration is open but unpursued.* The registration was never filed; §2 explains why it could not honestly have been. What stands is the falsifiable structure — prediction, kill thresholds, runtime verification — not a live claim.

- *Record 30's $\Omega_\Lambda = 1/\sqrt{2}$ is withdrawn as a value claim.* It was circular: the vacuum offset V_0 was tuned to produce it. (A chiral *ceiling* $\Omega \leq 1/\sqrt{2}$ is a separate bound and is not adjudicated here.)
- *The Ω contradiction ($1/\sqrt{2}$ vs $\ln 2$) is open, not resolved.* No reconciliation is offered and none is invented. The $\ln 2$ value remains what the script shows: a falsifiable open structure with named kill criteria, nothing more.

A Verification script: key excerpts

The full script, `verify_omegacdm_prediction.py`, ships alongside this folio and is the authoritative version. Excerpts below; every printed verdict is derived from a comparison the process performs, and any uncomputable check prints `UNCOMPUTED` rather than PASS.

```
# Inputs are published values, cited inline.
PLANCK_OL, PLANCK_OL_SIG = 0.6847, 0.0073 # Planck 2018 VI
PLANCK_OM, PLANCK_OM_SIG = 0.3153, 0.0073
PLANCK_H0 = 67.36
OMEGA_M_H2 = 0.1430
SHOES_H0, SHOES_H0_SIG = 73.04, 1.04 # Riess et al. 2022

def check(label, computed, expected, tol, unit=""):
   """Record a comparison. Verdict is computed, never asserted."""
   ok = abs(computed - expected) <= tol
   checks.append(ok)
   ...
```

```
# P3 - the directional bet, and the kill threshold
for sig, src in [(0.0073, "Planck 2018"), (0.0040, "DESI+Euclid"),
             (0.0028, "CMB-S4")]:
   lo, hi = LN2 - 3 * sig, LN2 + 3 * sig
   n = abs(LN2 - PLANCK_OL) / sig
   verdict = "FALSIFIED" if n > 3 else "alive"
```

```
# the internal tension, stated against our own interest
OL_at_shoes = 1.0 - OMEGA_M_H2 / (SHOES_H0 / 100.0) ** 2
n = abs(OL_at_shoes - LN2) / PLANCK_OL_SIG
# -> Omega_L = ln2 needs H_0 ~ 68.3;
# a_0 = cH_0/2pi prefers H_0 ~ 73.04.
# NOT jointly satisfiable. A converged H_0 kills at least one.
```

Run output (exit 0; 6/6 checks agree): $H_0 = 68.27$, $\Omega_m = 0.306853$, Planck offset $+1.16\sigma$, SH0ES tension -4.59σ, kill at $\sigma(\Omega_\Lambda) = 0.0028$, SH0ES-implied $\Omega_\Lambda = 0.7320$ at 5.3σ from $\ln 2$, epoch window $-0.0262 \leq z \leq +0.0001$.

References

[1] Planck Collaboration, "Planck 2018 results. VI. Cosmological parameters," *Astron. Astrophys.* **641**, A6 (2020).

[2] A. G. Riess et al., "A comprehensive measurement of the local value of the Hubble constant with $1\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$ uncertainty," *Astrophys. J. Lett.* **934**, L7 (2022).

[3] R. W. Yett, "ΩCDM: A Pre-Registered Cosmological Prediction: $\Omega_\Lambda = \ln 2$," Zenodo `10.5281/zenodo.21867985` (2026). Superseded by this folio; retained as history.

[4] Record 30 of the Zenodo corpus ($\Omega_\Lambda = 1/\sqrt{2}$ via tuned V_0): value claim **withdrawn** as circular by the status note of this folio (§4).

[5] R. W. Yett and Chyren, "The Law of GOD: A Pleroma of Geometrically Ordered Dynamics — Axioms, Operator, and Open Obligations," IT-PLEROMA successor folio (this series, Paper A). The axiomatic angle on the same central claim.